

EDA report – Automobile dataset

Exploratory data analysis | 2014 automobile dataset

Author: Lerato Kgosimore

1. Introduction

Exploratory data analysis (EDA) is a foundational step in any data science workflow. It allows us to understand the structure and quality of a dataset before drawing conclusions or building predictive models. This report presents a full EDA of the automobile dataset, which contains specifications and pricing for 205 vehicle models from a range of manufacturers.

The analysis addresses the following questions:

- Which are the five most expensive cars?
- Which manufacturer builds the most fuel-efficient vehicles?
- Which vehicles have the largest engine capacity?
- Which manufacturer has the most car models in the dataset?
- Is there a relationship between engine size and price?
- How are body styles distributed across the dataset?

2. Data cleaning

2.1 Initial data assessment

The raw dataset contained **205 rows** and **26 columns**. Several data quality issues were identified during initial inspection:

- Missing values encoded as '?' rather than NaN across multiple columns (e.g. horsepower, bore, stroke, peak-rpm, price).
- Unnecessary columns: 'symboling' and 'normalized-losses' were not relevant to the analysis.
- Numeric columns stored as object (text) data types, preventing mathematical operations.
- Potential duplicate rows.

2.2 Cleaning steps applied

Step	Action	Result
1. Drop columns	Removed 'symboling' and 'normalized-losses'	24 columns remaining
2. Replace '?'	Replaced all '?' values with NaN	33 missing values identified
3. Remove duplicates	Dropped all duplicate rows	0 duplicate(s) removed
4. Type conversion	Converted 14 numeric columns from object to float64	Types corrected
5. Drop missing rows	Discarded any row with at least one NaN	31 rows removed → 174 rows retained

After cleaning, the dataset contained **174 rows** and **24 columns**, ready for analysis.

3. Analysis and visualisations

3.1 Which are the five most expensive cars?

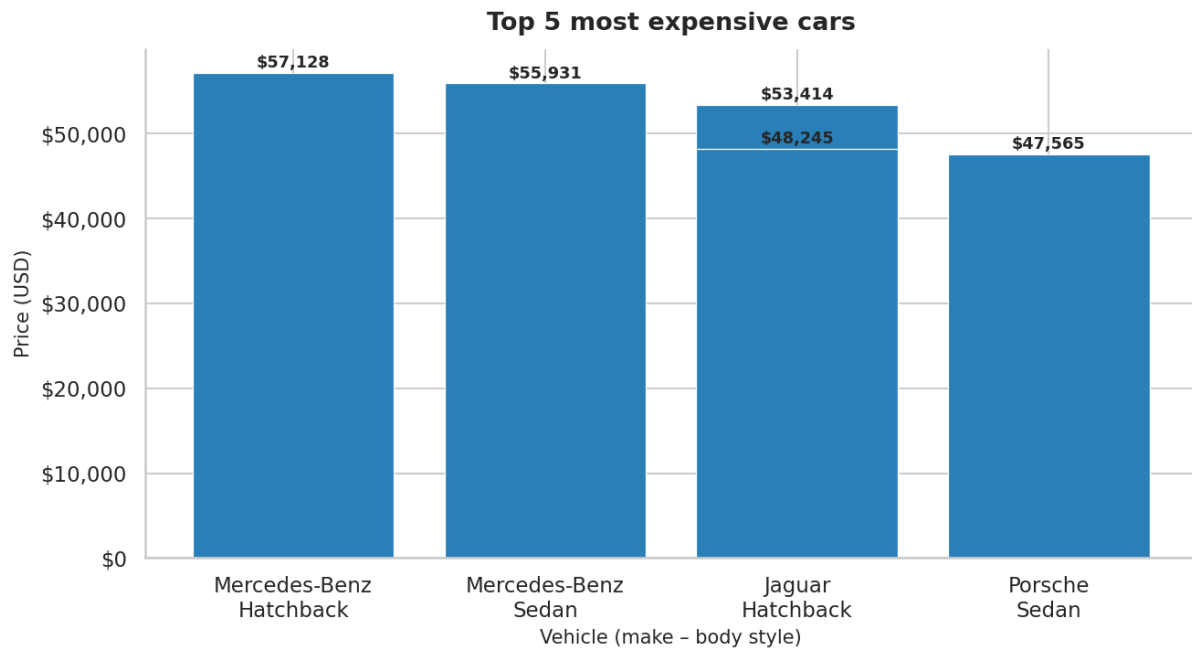


Figure 1: Bar chart showing the five highest-priced vehicles in the dataset, sorted in descending order by price.

The five most expensive vehicles in the dataset are all from premium manufacturers. The highest-priced model is a **Mercedes-Benz hatchback** at **\$57,128**. Premium brands such as Mercedes-Benz, BMW, and Porsche feature prominently, reflecting the premium placed on luxury specifications such as larger engines and refined body styles. Price in this segment is strongly influenced by engine size and brand prestige.

3.2 Which manufacturer builds the most fuel-efficient vehicles?

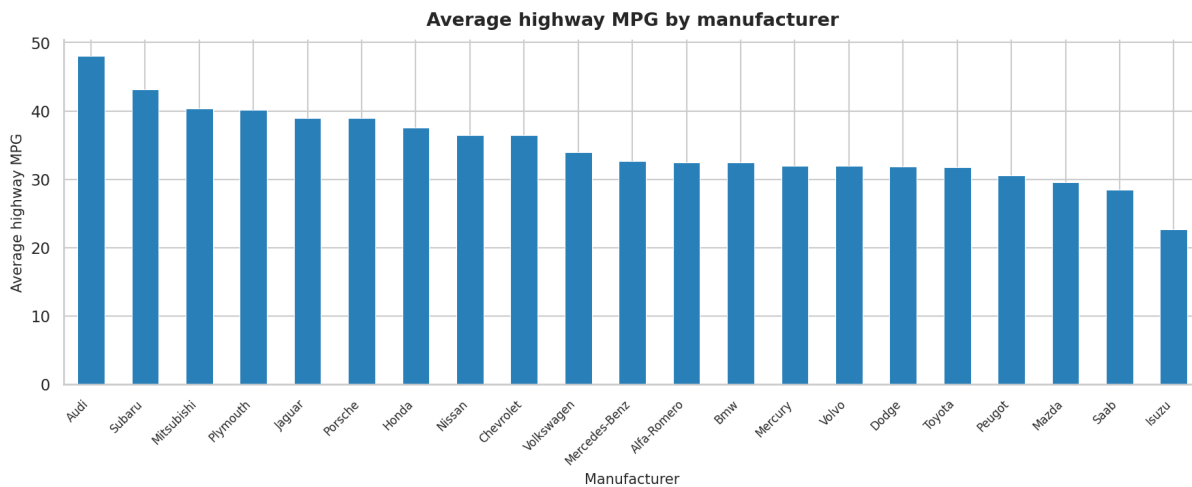


Figure 2: Average highway MPG for each manufacturer, sorted from most to least efficient.

Audi leads all manufacturers with an average highway fuel economy of **48.0 MPG**. Smaller, lighter manufacturers generally outperform larger ones in fuel efficiency. **Isuzu** has the lowest average MPG, likely due to heavier, higher-powered models in their lineup. This chart clearly shows that fuel efficiency varies widely across manufacturers and is not solely determined by brand origin.

3.3 Which vehicles have the largest engine capacity?

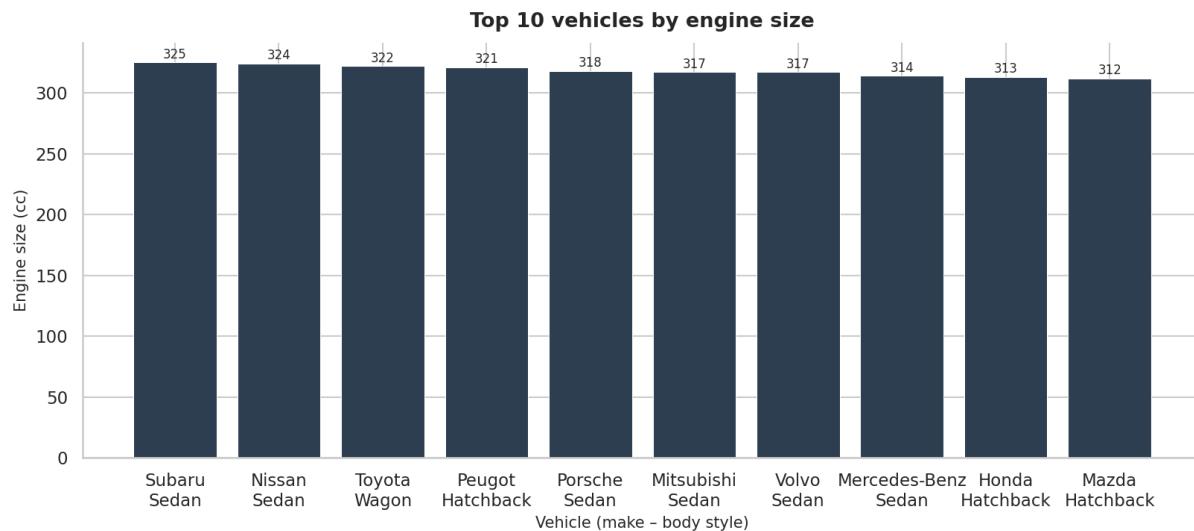


Figure 3: Top 10 vehicles ranked by engine displacement (cc).

The vehicle with the largest engine is a **Subaru sedan** with a displacement of **325 cc**. Large engine capacities are concentrated in luxury and sports brands. Vehicles with larger engines generally also carry higher price tags, as seen in the previous analysis. The number of cylinders is closely correlated with engine size.

3.4 Which manufacturer has the most car models in the dataset?

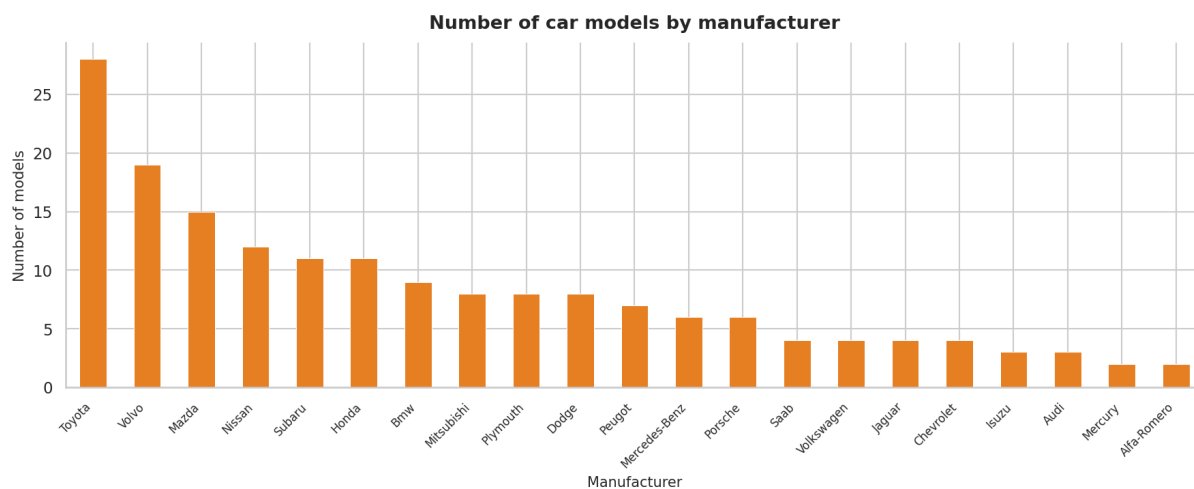


Figure 4: Count of car models per manufacturer in the dataset.

Toyota has the greatest representation in the dataset with **28 models**, followed by several European and American manufacturers. The dataset is unbalanced — a few manufacturers contribute many models while others appear only once or twice. This should be considered when drawing manufacturer-level conclusions, as averages for under-represented makes may be less reliable.

3.5 Is there a relationship between engine size and price?

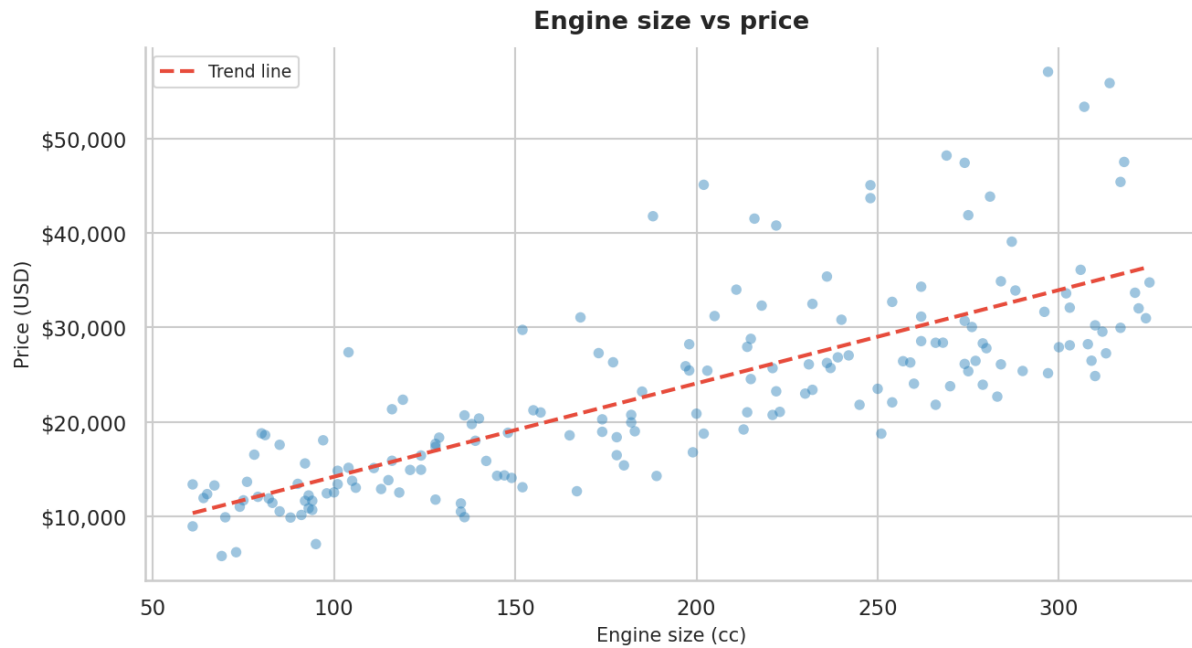


Figure 5: Scatter plot of engine size (cc) against price (USD), with a linear trend line.

There is a **positive correlation (0.77)** between engine size and price. As engine displacement increases, vehicle price tends to rise as well. The trend line confirms this relationship, although there is considerable spread around the line, meaning engine size alone does not fully explain price variation — other factors such as brand, body style, and added features also play a role.

3.6 How are body styles distributed across the dataset?

Distribution of body styles

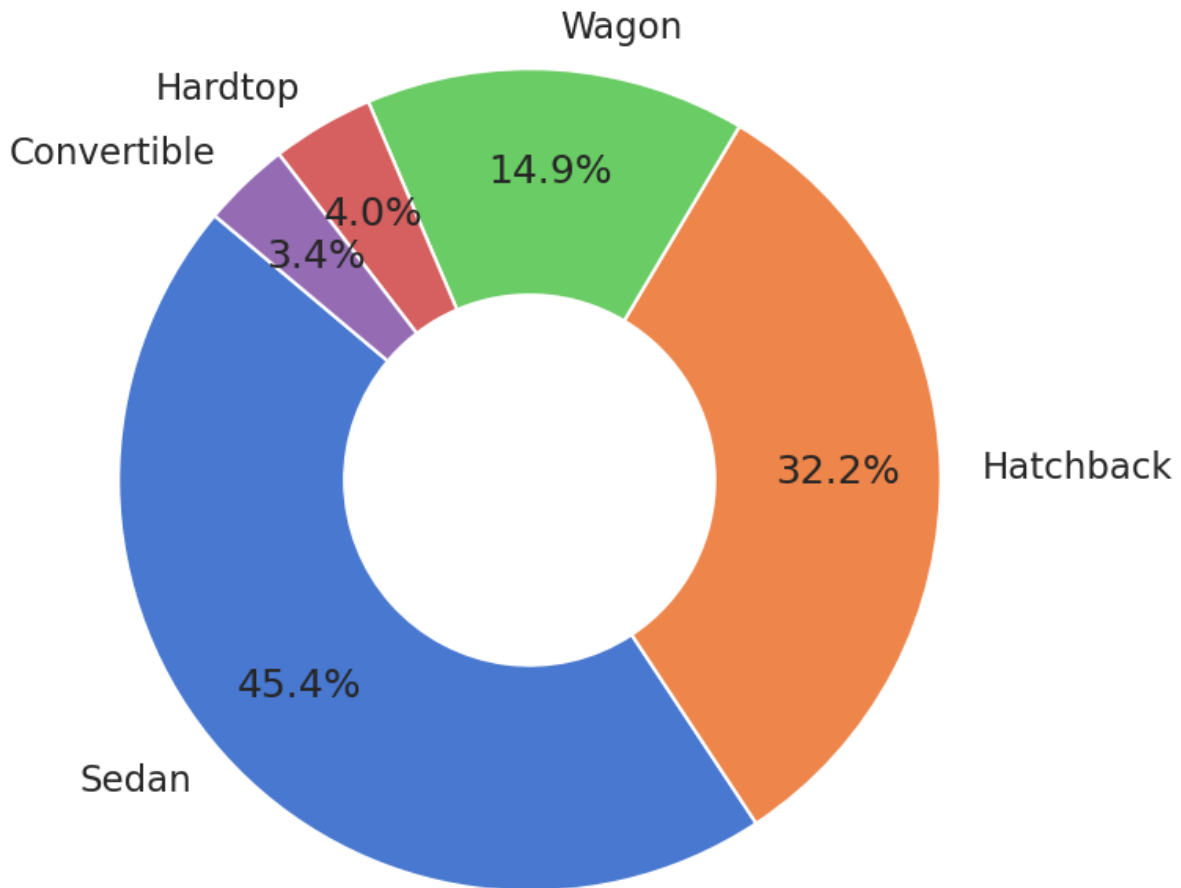


Figure 6: Donut chart showing the proportion of each body style in the dataset.

Sedan is the most common body style, accounting for approximately **45.4%** of all vehicles. Sedans and hatchbacks together make up the majority of the dataset, reflecting typical consumer preferences in the mid-1980s passenger car market. Convertibles and hardtops are the least represented styles.

4. Conclusions

This exploratory data analysis of the automobile dataset revealed several meaningful insights about vehicle specifications, pricing, and manufacturer characteristics:

- **Price is driven by brand and engine size.** Luxury manufacturers consistently command the highest prices, and a positive correlation exists between engine displacement and price.
- **Fuel efficiency varies widely by manufacturer.** Smaller, lighter-focused brands outperform larger ones in highway MPG, while manufacturers with powerful lineups trail significantly.
- **Toyota dominates in model count.** The dataset is manufacturer-unbalanced, with Toyota representing the largest share of entries.
- **Sedans and hatchbacks dominate the market.** Together they account for the majority of body styles, reflecting the mainstream passenger car market of the era.
- **Data quality required careful attention.** Missing values encoded as '?' and incorrect data types required systematic cleaning before any reliable analysis could be performed.

Overall, this EDA provides a solid foundation for further analysis, such as building a predictive pricing model or segmenting vehicles by performance and efficiency profiles.